

Toothbrush Holder

3D Printed Parts

PLA

1x Main body This is the main housing for all the parts it also has two tabs that help lock the toothbrush in place.

2x Side doors This shields the toothbrush from debris and stops UVC light from getting out.

2x Rods Helps lock the toothbrush in and helps with the movement of the doors.

TPU

1x Band Helps to grip the toothbrush and allows compatibility with most standard toothbrushes.

Electronics

Microcontroller

1x Arduino I will use this for controlling the servos and other output as well as sensing input

Effectors

1x UVC LED(using a blue led instead for Standin) This is for sterilizing the toothbrush.

2x Servo For locking the side doors in the closed position.

1x Green LED For letting the user know the UVC led is on.

1x Red LED for letting the user know something is wrong.

1x Active Buzzer for letting user know the UVC sterilization is done

Sensors

2x Buttons one For sensing when a toothbrush is inserted. The other one is for turning on the sterilization process.

1x photoresistor for detecting if the door is closed.

Power source

Lithium-ion rechargeable battery(will just plug in for now) for powering the circuit